# Lecture 23: Integer Optimization

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# Announcements

- Homework 8 due today
- Homework 9 posted, due on May 10th at noon (12pm)
- Two more lectures on May 5 and May 6, respectively

Regarding final exam:

- Time: May 17, 8:30am-10:30am. The location is TBD.
- You can bring two pieces of A4 papers (double sided). No other materials are permitted (no electronic device including calculator).
- A sample exam and its solution are posted.
- A review slides will be posted later.

# Recap: Nonlinear Optimization

Optimality conditions

- KKT conditions
- Be able to formulate and analyze KKT conditions

Convexity

- Know the concepts: convex set, convex/concave function, convex constraints, and convex optimization problem
- Can identify them
- Can convert problems into convex optimization problem (if possible)

# Recap: Nonlinear Optimization

Algorithms

- Bisection, Golden section method
- Gradient descent, Newton's method
- Projected gradient method
- Understand the ideas and procedures, can write some codes to implement them

# Agenda

What we have studied so far are all continuous optimization problems (both LP and NLP)

- Decision variables can take continuous values (any fractional values)

In the remaining lectures, we are going to study discrete optimization problems

- Decision variables can only take discrete values
- We mainly consider integer linear programs

# Integer Linear Program

An integer linear program (IP) is a linear program with the additional constraint that all variables must be integers:

$$\begin{aligned} \text{minimize}_{\mathbf{x}} \quad & \mathbf{c}^T\mathbf{x} \\ \text{subject to} \quad & A\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq 0 \\ & \mathbf{x} \in \mathbb{Z}^n \end{aligned}$$

Here we use $\mathbb{Z}$ to denote the set of integers.

- One may also encounter mixed integer programs (MIP), in which one set of variables must be integer and the rest are allowed to be continuous

# A Special Case: Binary IP

One subclass of IP is binary integer programs, in which each decision variable has to be either 0 or 1:

$$\begin{aligned} \text{minimize}_{\mathbf{x}} \quad & \mathbf{c}^T\mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} = \mathbf{b} \\ & x_i \in \{0, 1\} \end{aligned}$$

- The variables in a binary IP are also called indicator variables

# Integer Programs

In many real applications, the decision variables should be integers (or binary)

- Production planning
- Nurse scheduling
- Transportation problem
- Shortest path
- Vertex cover

In the previous discussions, we ignored those integral constraints for the ease of analysis. However, it is important to study such problems since they do capture many practical situations

# Example: Knapsack Problem

John is planning for a trip. There are $n$ items he would like to bring with him.

- The $i$th item has value $v_i$
- The weight of $i$th item is $a_i$
- His bag has a maximum allowable weight $C$
- He wants to bring as much value as possible

Decision variables:

- $x_i$: whether to bring $i$th item or not. $x_i \in \{0, 1\}$

Optimization problem:

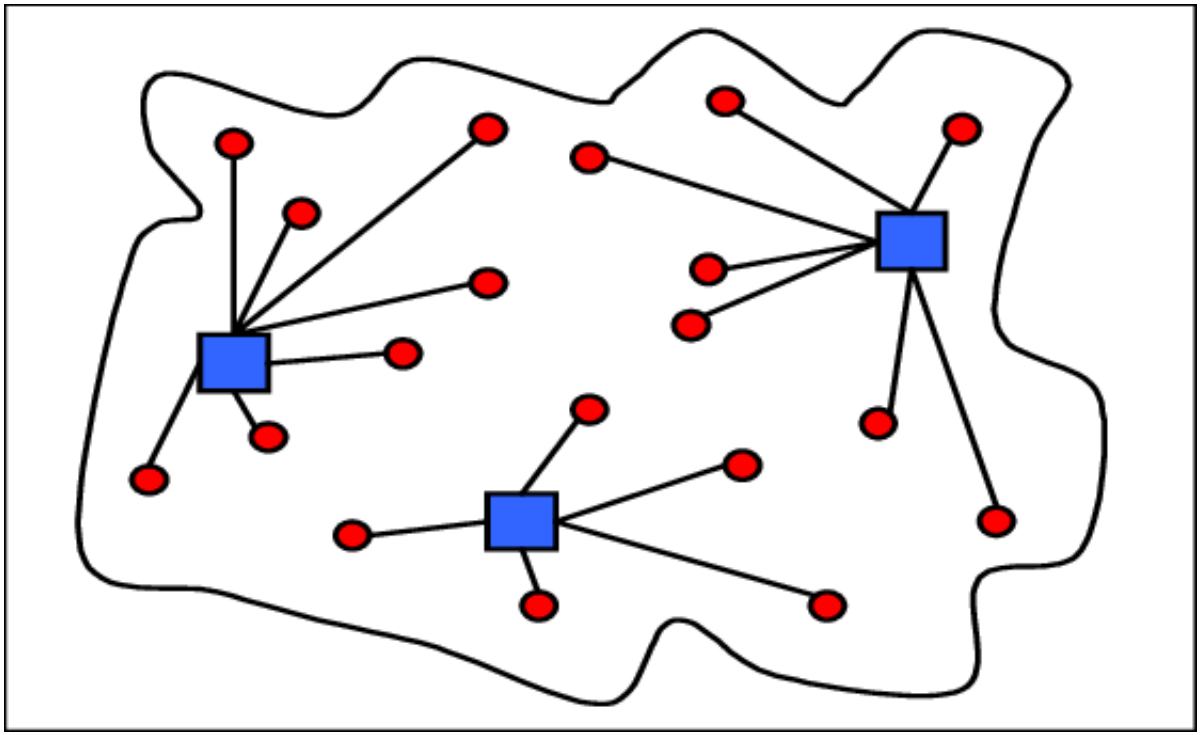
$$\begin{aligned} \text{maximize}_{\mathbf{x}} \quad & \sum_{i=1}^{n} v_i x_i \\ \text{subject to} \quad & \sum_{i=1}^{n} a_i x_i \leq C \\ & x_i \in \{0, 1\} \end{aligned}$$

It is a binary optimization problem

# Example: Warehouse/Facility Location Problem

- There are $n$ warehouses available for use
- Opening warehouse $i$ has a fixed operating cost $f_i$
- There are $m$ customers
- Customer $j$ has demand $d_j$. It costs $c_{ij}$ to ship one unit of product from warehouse $i$ to customer $j$
- The objective is to satisfy all customers' demands, while minimizing the total costs (operating + shipment)

# Facility Location Problem



# Formulation

Decision variables:

- $y_i$: whether to open warehouse $i$ or not ($y_i \in \{0, 1\}$)
- $x_{ij}$: how many units to ship from warehouse $i$ to customer $j$

Objective function:

$$\text{minimize}_{\mathbf{x},\mathbf{y}} \quad \sum_{i=1}^{n} f_i y_i + \sum_{i,j} c_{ij} x_{ij}$$

Constraints:

- Need to satisfy every customer's demand: $\sum_{i=1}^{n} x_{ij} \geq d_j$
- A location can provide supply only if it is opened
- To capture this, we can define constraints (why):

$$x_{ij} \leq d_j y_i$$

# Formulation

Therefore we can formulate the warehouse/facility location problem as follows:

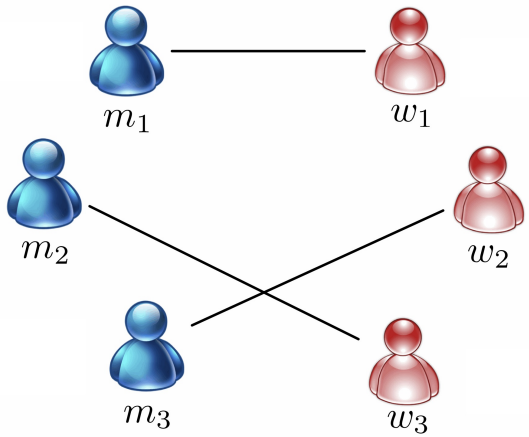
$$
\begin{aligned}
\text{minimize}_{\mathbf{x},\mathbf{y}} \quad & \textstyle\sum_{i=1}^{n} f_i y_i + \sum_{i,j} c_{ij} x_{ij} & \\
\text{subject to} \quad & \textstyle\sum_{i=1}^{n} x_{ij} \geq d_j, & \text{for all } j \\
& x_{ij} \leq d_j y_i, & \text{for all } i, j \\
& x_{ij} \geq 0 & \\
& y_i \in \{0, 1\} &
\end{aligned}
$$

It is a mixed integer program

# Example: Matching Problem

Assume there are $n$ girls and $n$ boys' information in a dating website.

- Each girl $i$ has rated boy $j$ with a score $v_{ij}$
- The website wants to match one girl with one boy so as to maximize the total score of the matching



# Matching Problem

Decision variables:

- $x_{ij}$: whether girl $i$ is matched with boy $j$

Objective function

$$\text{maximize} \quad \sum_{i=1}^{n} \sum_{j=1}^{n} v_{ij} x_{ij}$$

Constraints:

- Each girl is matched to exactly one boy:

$$\sum_{j=1}^{n} x_{ij} = 1$$

- Each boy is matched to exactly one girl:

$$\sum_{i=1}^{n} x_{ij} = 1$$

- The assignment must be integers: $x_{ij} \in \{0, 1\}$

# Matching Problem: Formulation

$$\begin{aligned}
\text{maximize} \quad & \sum_{i=1}^{n}\sum_{j=1}^{n} v_{ij}x_{ij} & \\
\text{s.t.} \quad & \sum_{i=1}^{n} x_{ij} = 1 & \forall j \\
& \sum_{j=1}^{n} x_{ij} = 1 & \forall i \\
& x_{ij} \in \{0,1\} &
\end{aligned}$$

It is a binary optimization problem

- It is also used in e.g., search engine adwords allocation problem, object tracking, and many other useful problems

# Using Binary Variable to Capture Certain Constraints

Suppose we want to incorporate a constraint $|x| \geq 1$ in an optimization problem (equivalent to $x \geq 1$ or $x \leq -1$)

- Cannot be captured in a single linear constraint
- It is not a convex constraint
- However, we can introduce a binary variable $y$ and write the constraint as

$$\begin{aligned} x &\geq 1 - My; \\ x &\leq -1 + M(1-y); \\ y &\in \{0, 1\} \end{aligned}$$

  where $M$ is a very big number
- This will be a set of linear constraints with a binary variable

# Our Objective

We have seen that integer optimization problems arise naturally from many applications.

We want to answer two questions in our study:

- How different is it from linear optimization problems?
- How to solve IPs? Any interesting properties?

We can give a quick answer to the first question:

- It is very different
- Remember we have defined the complexity class NP-Hard to denote a class of hard problems. Integer optimization is NP-Hard, thus unlikely to have polynomial-time algorithm
- Even to verify if a set (polyhedron) contains an integral point is very hard

# An Intuitive Thought

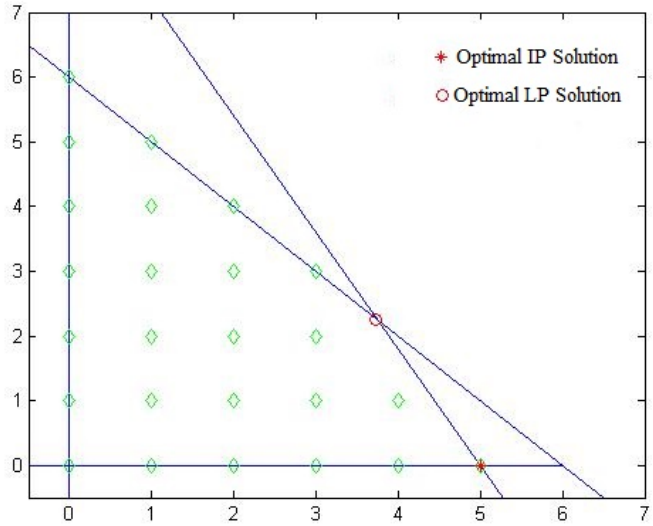
Maybe we can first relax the integral constraints, solve the linear program (which is easy) and find some integral solution near the optimal solution to the linear program.

- We call the corresponding linear program the *LP relaxation* of the IP
- Definitely a plausible method
- May work well in some case
- But may fail in other cases

# An Example

Consider the following integer program (a production planning problem):

$$\begin{aligned} \text{maximize} \quad & 8x_1 + 5x_2 \\ \text{subject to} \quad & 9x_1 + 5x_2 \leq 45 \\ & x_1 + x_2 \leq 6 \\ & x_1, x_2 \geq 0 \\ & x_1, x_2 \in \mathbb{Z} \end{aligned}$$




- If we consider the LP relaxation, we get the optimal solution is $(x_1, x_2) = (15/4, 9/4)$.
- If we try to round it to the nearest integral solution, we get $(4, 2)$ which is not even feasible for the integer program
- If we consider the nearest feasible integral solution, it will be $(3, 2)$, the objective value is 34
- The optimal solution to the integer program is actually $(5, 0)$, the objective value is 40

# Another Example: Knapsack Problem

There are three items:

- Item A: 3kg with value 300
- Item B: 2kg with value 180
- Item C: 2kg with value 170

The capacity of the bag is 4kg.

- The relaxation of a binary constraint $x \in \{0, 1\}$ is $x \in [0, 1]$.
- Optimal solution to LP relaxation: (1, 1/2, 0) with optimal value 390
- Nearest feasible integral solution: (1, 0, 0) with value 300
- Optimal integral solution: (0, 1, 1) with value 350

# Summary

The optimal solution to the integer program might not be close to the optimal solution to the LP relaxation.

- Directly round the LP solution to integers may not get one a good solution
- The reason is the very combinatorial nature of the IP problem (no continuity in the solution space).

But does the LP relaxation provide any useful information?

# LP Relaxation as a Bound for IP

## Theorem (LP Relaxation as a Bound for IP)

1. *For a maximization problem, the optimal value of the LP relaxation provides an upper bound for the optimal value of the IP.*
2. *For a minimization problem, the optimal value of the LP relaxation provides a lower bound for the optimal value of the IP.*

The difference between the optimal value of the LP and the IP is called the *integrality gap*

- For maximization problems, the integrality gap is $v^{LP} - v^{IP}$
- For minimization problems, the integrality gap is $v^{IP} - v^{LP}$

# Use LP Relaxation as a Bound

Consider a maximization problem, suppose we solved the LP relaxation and get the optimal value is $v^{LP}$ (with some fractional solution).

Then we use some method to obtain a feasible integral solution (e.g., by rounding the solution to the LP relaxation) and find the objective value is $v^0$.

Then the difference between the obtained solution and the optimal integral solution (whose objective value is $v^{IP}$) satisfies:

$$0 \leq v^{IP} - v^0 \leq v^{LP} - v^0$$

That is, one can use the LP relaxation solution to construct a bound on how good a certain integral solution is.

# Some Good Cases

There are some cases where solving the LP relaxation can give one very good solution

**Theorem**

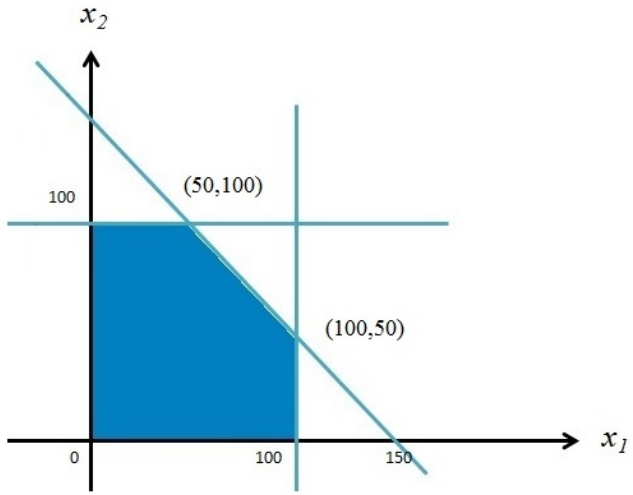
*If the optimal solution to the LP relaxation is integral, then the solution must be optimal to the IP problem.*

# When Does LP Have Integral Solutions

Remember for LP, there must exist an optimal solution that is a basic feasible solution.

- If we can guarantee that every BFS is integral, then the LP must have an integer optimal solution

For example, in the following production planning problem:




All the extreme points are integral. Therefore, there must exist an optimal solution that is integral.

- This criterion has nothing to do with the objective function
- Some sufficient conditions for this — totally unimodularity